

Sign analysis ,

Sign of	$0 < r < a$	$a < r < \infty$
$\frac{dS}{dr} = 4\pi r - \frac{2000}{r^2}$	(-)	(+)

S is $\searrow$

S is $\nearrow$

Therefore, there is a minimum at ,

$$r = \left(\frac{500}{\pi} \right)^{1/3}$$

$$\text{From } h = \frac{1000}{\pi r^2} ,$$

$$h = 2 \left(\frac{500}{\pi} \right)^{1/3}$$

Integration

(Reverse)

$$F(x) = x^2$$

$$F'(x) = 2x$$

$$f(x) = 2x$$

$$\frac{d}{dx} (\underbrace{f(x)}) = \underbrace{f(x)}$$

derivative of $f(x)$

anti derivative
of $f(x)$

$x^2 + c$; c is an arbitrary constant. is the antiderivative of $2x$

$$\int f(x) dx = f(x) + c$$

$$\int \underbrace{2x}_{\text{derivative}} dx = \underbrace{x^2 + c}_{\text{integral}} ; c \text{ is an arbitrary constant.}$$

↑
integral sign

e.g: $\frac{d}{dx} x^3 = 3x^2$

$$\int 3x^2 dx = x^3 + C$$

$$\frac{d}{dx} (x^3 + 2) = 3x^2$$

$$\frac{d}{dx} (x^3 + 15) = 3x^2$$

✓ $\int 3x^2 dx = x^3 + C$; C is an arb. const.

e.g : $\frac{d}{dx} \sin x = \cos x$

~~$$\int \cos x dx$$~~

$$\int \cos x dx = \sin x + C$$

$$\frac{d}{dx} \sin x + \frac{1}{2} = \cos x$$

$$\int \cos x dx = \sin x + C ;$$

e.g : $\frac{d}{dx} e^x = e^x$

$$\int e^x dx = e^x + C ;$$

✓	$\frac{d}{dx} x^n = nx^{n-1}$
✓	$\int nx^{n-1} dx = x^n + C$

General Formula

$\int x^n dx = \frac{x^{n+1}}{n+1} + C$	$n \neq -1$	$\int x^{-1} dx = \ln x + C$
*	• special	*

Example

$$\textcircled{i)} \int x^{20} dx = \frac{x^{20+1}}{20+1} + C$$

$$= \frac{x^{21}}{21} + C ; C \text{ is an arb. const.}$$

$$\textcircled{ii)} \int x^{-1} dx = \int \frac{1}{x} dx = \ln|x| + C ; C \text{ is an arb. const.}$$

$$\textcircled{iii)} \int \frac{1}{x^2} dx = \int x^{-2} dx = \frac{x^{-2+1}}{-2+1} + C$$

$$= \frac{x^{-1}}{-1} + C ;$$

$$\textcircled{iv)} \int \frac{1}{x^8} dx = \int x^{-8} dx = \frac{x^{-8+1}}{-8+1} + C$$

$$= \frac{x^{-7}}{-7} + C ;$$

* Sum Rule for Integrals.

$$\boxed{\int [f(x) + g(x)] dx = \int f(x) dx + \int g(x) dx}$$

* Constant Multiple Rule for integrals.

$$\boxed{\int c \cdot f(x) dx = c \cdot \int f(x) dx}$$

example $\rightarrow \int 3x^2 dx = 3 \int x^2 dx$

$$= 3 \times \frac{x^{2+1}}{2+1} + C$$

$$= 3 \times \frac{x^3}{3} + C$$

$$= \underline{\underline{x^3 + C}}; \text{ } C \text{ is an arb. const.}$$

example $\rightarrow \int (3x^3 - x^4 + 5x^2 - 1) dx$

$$= \int 3x^3 dx - \int x^4 dx + \int 5x^2 dx - \int 1 dx$$

$$= 3 \int x^3 dx - \int x^4 dx + 5 \int x^2 dx - \int 1 dx$$

$$= 3 \times \frac{x^{3+1}}{3+1} - \frac{x^{4+1}}{4+1} + 5 \times \frac{x^{2+1}}{2+1} - x + C$$

$$= 3 \times \frac{x^4}{4} - \frac{x^5}{5} + 5 \times \frac{x^3}{3} - x + C ;$$

Example

② d) $\int (x^2 + 3x - 1) dx =$

$$= \int x^2 dx + \int 3x dx - \int 1 dx$$

$$= \int x^2 dx + 3 \int x dx - \int 1 dx$$

$$= \frac{x^{2+1}}{2+1} + 3 \times \frac{x^{1+1}}{1+1} - x + C$$

$$= \frac{x^2}{3} + \frac{3x^2}{2} - x + C ;$$

$$(ii) \int \sqrt{x} + \frac{3}{x} dx = \int \sqrt{x} dx + 3 \int \frac{1}{x} dx$$

$$= \int x^{1/2} dx + 3 \int \frac{1}{x} dx$$

$$= \frac{x^{1/2+1}}{1/2+1} + 3 \ln |x| + C$$

$$= \frac{2x^{3/2}}{3} + 3 \ln |x| + C ;$$

$$(iii) \int \left(x + \frac{1}{x}\right)^2 dx = \int x^2 + 2x \cdot x \cdot \frac{1}{x} + \left(\frac{1}{x}\right)^2 dx$$

$$= \int x^2 + 2 + \frac{1}{x^2} dx$$

$$= \int x^2 dx + 2 \int dx + \int \frac{1}{x^2} dx$$

$$= \frac{x^3}{3} + 2x + \int x^{-2} dx$$

$$= \frac{x^3}{3} + 2x + \frac{x^{-2+1}}{-2+1} + C$$

$$= \frac{x^3}{3} + 2x - \frac{1}{x} + C ; C \text{ is an arb. const.}$$

$$\begin{aligned}
 \text{(iv)} \quad \int \left(\frac{3x}{100} + \frac{4}{x^4} \right) dx &= \int \frac{3x}{100} dx + \int \frac{4}{x^4} dx \\
 &= \frac{3}{100} \int x dx + 4 \int x^{-4} dx \\
 &= \frac{3}{100} \left(\frac{x^{1+1}}{1+1} \right) + 4 \times \frac{x^{-4+1}}{-4+1} + C \\
 &= \frac{3x^2}{200} + \frac{4}{(-3x^3)} + C \\
 &= \frac{3x^2}{200} - \frac{4}{3x^3} + C ;
 \end{aligned}$$

$$\begin{aligned}
 \text{(v)} \quad \int \left(\frac{3 + 5x - 6x^2 - 7x^3}{2x^2} \right) dx &= \int \left(\frac{3}{2x^2} + \frac{5x}{2x^2} - \frac{6x^2}{2x^2} - \frac{7x^3}{2x^2} \right) dx \\
 &= \int \frac{3}{2x^2} dx + \int \frac{5}{2x} dx \\
 &\quad - \int 3 dx - \frac{7}{2} \int x dx \\
 &= \frac{3}{2} \int x^{-2} dx + \frac{5}{2} \int \frac{1}{x} dx - 3 \int dx - \frac{7}{2} \int x dx \\
 &= \frac{3}{2} \left(\frac{x^{-2+1}}{-2+1} \right) + \frac{5}{2} (\ln |x|) - 3x - \frac{7}{2} \cdot \frac{x^{1+1}}{1+1} \\
 &= \frac{3}{2} \times \frac{x^{-1}}{-1} + \frac{5}{2} \ln |x| - 3x - \frac{7x^2}{4} + C \\
 &= -\frac{3}{2} \left(\frac{1}{x} \right) + \frac{5}{2} \ln |x| - 3x - \frac{7x^2}{4} + C ;
 \end{aligned}$$

* Integral of e^x

$$\int e^x dx = e^x + C$$

* Integrals of Basic Trigonometric Functions.

$$\int \sin x dx = -\cos x + C$$

$$\int \cos x dx = \sin x + C$$

$$\int \sec^2 x dx = \tan x + C$$

$$\int \csc^2 x dx = -\cot x + C$$

$$\int \sec x \tan x dx = \sec x + C$$

$$\int \csc x \cot x dx = -\csc x + C$$

Example

③ (i)

$$\begin{aligned}\int (e^x + \sin x) dx &= \int e^x dx + \int \sin x dx \\ &= \cancel{e^x + C} - \cos x + C \\ &= e^x - \cos x + C ; \text{ C is arb. const.}\end{aligned}$$

$$\begin{aligned}\text{(ii)} \quad \int 3e^x dx &= \int 3e^x dx \\ &= 3 \int e^x dx \\ &= 3e^x + C ;\end{aligned}$$

$$\begin{aligned}\text{(iii)} \quad \int (\cos x + 5) dx &= \int \cos x dx + \int 5 dx \\ &= \sin x + 5x + C ;\end{aligned}$$

$$\begin{aligned}\text{(iv)} \quad \int (2 \sin x - 4e^x) dx &= \int 2 \sin x dx - \int 4e^x dx \\ &= 2x - \cos x - 4xe^x + C \\ &= -2\cos x - 4e^x + C ;\end{aligned}$$

$$(v) \int (7 \cos x + 3e^x) dx =$$

$$\begin{aligned} & \cancel{\int 7 \cos x} + \int 7 \cos x dx + \int 3e^x dx \\ &= 7x \sin x + 3xe^x + C \\ &= 7 \sin x + 3e^x + C ; \end{aligned}$$

$$(vi) \int 5(e^x - \sin x) dx =$$

$$\begin{aligned} & \int 5e^x dx - \int 5 \sin x dx \\ & 5xe^x - 5x - \cos x + C \\ & 5e^x + 5 \cos x + C ; \end{aligned}$$

$$(vii) \int (x^2 + 3)^2 dx$$

$$\cancel{= \int 2(x^2 + 3) dx}$$

$$\cancel{= \int 2x^2 dx + \int 2 \times 3 dx}$$

$$\cancel{= 2}$$

$$\rightarrow = \int ((x^2)^2 + 2 \times x^2 \times 3 + 3^2) dx$$

$$= \int (x^4 + 6x^2 + 9) dx$$

$$= \int x^4 dx + 6 \int x^2 dx + 9 \int dx$$

$$= \frac{x^5}{5} + \frac{6x^3}{3} + 9x + C ; \text{ C is an arb. const.}$$

$$(viii) \int (x - \frac{1}{x})^2 dx = \int ((x^2) - 2 \times x \times \frac{1}{x} + (\frac{1}{x})^2) dx$$

$$= \int (x^2 - 2 + \frac{1}{x^2}) dx$$

$$= \int x^2 dx - 2 \int 1 dx + \int x^{-2} dx$$

$$= \frac{x^3}{3} - 2x + \frac{x^{-1}}{-1} + C$$

$$= \frac{x^3}{3} - 2x - \frac{1}{x} + C ;$$

• Method of Substitution.

$$\int \underbrace{(3x+7)}_u^{20} dx$$

$u = 3x + 7$

$$\frac{du}{dx} = 3 \Rightarrow du = 3 dx$$

$$dx = \frac{1}{3} du$$

$$\int u^{20} \frac{du}{3} = \int \frac{1}{3} u^{20} du$$

$$= \frac{1}{3} \int u^{20} du$$

$$= \int u^{20} \frac{du}{3} = \frac{1}{3} \int u^{20} du$$

$$= \frac{1}{3} \cdot \frac{u^{21}}{21} + C ;$$

$$= \frac{u^{21}}{63} + C ;$$

→ Back substitute for u is obtain the answer is x ,

$$\int (3x+7)^{20} dx = \frac{(3x+7)^{21}}{63} + C ; \text{ } C \text{ is an arbitrary constant.}$$

Example

(i) $\int \frac{1}{2x-3} dx$

Substitution $u = 2x - 3$

differentiate $\frac{du}{dx} = 2 \Rightarrow dx = \frac{du}{2}$

Rewrite the problem : $\int \frac{1}{u} \frac{dx}{2} = \int \frac{1}{2u} du$

$$= \frac{1}{2} \int \frac{1}{u} du$$

$$= \frac{1}{2} \ln |u| + C$$

Back substitute :

$$= \frac{1}{2} \ln |2x-3| + C ; C \text{ is an arb. constant}$$

$$(ii) \int x(x^2+1)^{10} dx$$

$$\text{substitution } u = x^2 + 1$$

$$\text{Differentiate } \frac{du}{dx} = 2x \Rightarrow dx = \frac{du}{2x}$$

$$\text{Rewrite the problem } \int x \cdot u^{10} \cdot \frac{du}{2x}$$

$$= \int x \cdot u^{10} \cdot \frac{du}{2x}$$

$$= \int \frac{u^{10}}{2} du$$

$$\text{Solve the problem } \int \frac{u^{10}}{2} du = \frac{1}{2} \int u^{10} du$$

$$= \frac{1}{2} \times \frac{u^{11}}{11} + C$$

$$= \frac{u^{11}}{22} + C$$

$$\text{Back substitute : } \frac{(x^2+1)^{11}}{22} + C ;$$

(iii) $\int \sin 2x \, dx$

substitution $u = 2x$

differentiate $\frac{du}{dx} = 2 \Rightarrow dx = \frac{du}{2}$

Rewrite the problem $\int \sin u \frac{du}{2} = \frac{1}{2} \int \sin u \, du$

Solve the problem $\frac{1}{2} \int \sin u \, du = \frac{1}{2} (-\cos u) + C$

$$= \frac{1}{2} (-\cos 2x) + C$$

Back substitute : $\left(\frac{-\cos 2x}{2} \right) + C$; C is an arb. constant.

(iv) $\int \cos 5x \, dx$

substitution $u = 5x$

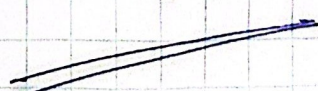
differentiate $\frac{du}{dx} = 5 \Rightarrow dx = \frac{du}{5}$

Rewrite the problem $\int \cos u \frac{du}{5} = \frac{1}{5} \int \cos u \, du$

Solve the problem $\frac{1}{5} \int \cos u \, du = \frac{1}{5} (\sin u) + C$

$$= \frac{1}{5} \sin 5x + C$$

Back substitute : $\frac{1}{5} \sin 5x + C$;



$$(v) \int \sin\left(2x - \frac{\pi}{10}\right) dx$$

substitution $u = 2x - \frac{\pi}{10}$

Differentiate $\frac{du}{dx} = 2 \Rightarrow dx = \frac{du}{2}$

~~Rewrite the problem~~

Rewrite the problem $\int \sin u \frac{du}{2} \Rightarrow \frac{1}{2} \int \sin u \cdot du$

$$= \frac{1}{2} (-\cos u) + C$$

Back substitute : $\frac{-\cos\left(2x - \frac{\pi}{10}\right)}{2} + C ;$

$$(vi) \int e^{2x} dx$$

substitution $u = e^{2x}$

differentiate $\frac{du}{dx} = 2 \Rightarrow dx = \frac{du}{2}$

Rewrite $\int e^u \frac{du}{2} = \frac{1}{2} \int e^u du$

$$= \frac{1}{2} e^u + C$$

Back substitute : $\frac{1}{2} e^{2x} + C ;$

$$(vii) \int x^2 \sqrt{x^3 + 5} \, dx$$

substitution $u = x^3 + 5$ $\frac{du}{dx} = \frac{1}{2}(x^3 + 5)^{-1/2} \cdot 3x^2$

differentiate $\frac{du}{dx} = 3x^2 \Rightarrow dx = \frac{du}{3x^2}$

Rewrite $\int x^2 \sqrt{u} \frac{du}{3x^2} = \frac{1}{3} \int \sqrt{u} \, du$

Solve : $\frac{1}{3} \int u^{1/2} \, du$

$$= \frac{1}{3} \cdot \frac{u^{3/2}}{3/2} + C$$

$$= \frac{2}{9} u^{3/2} + C$$

Back substitute : $\frac{2}{9} (x^3 + 5)^{3/2} + C$; C is an arb constant.

$$(viii) \int \frac{x}{x^2 - 1} \, dx$$

substitution $u = x^2 - 1$

differentiate $\frac{du}{dx} = 2x \Rightarrow dx = \frac{du}{2x}$

Rewrite $\int \frac{x}{u} \frac{du}{2x} = \int \frac{1}{2u} \, du$

$$= \frac{1}{2} \int \frac{1}{u} \, du$$

$$= \frac{1}{2} \ln |u| + C$$

$$= \frac{1}{2} \ln |x^2 - 1| + C ;$$

$$(ix) \int x e^{x^2} dx$$

Substitute $u = x^2$

$$\text{differentiate } \frac{du}{dx} = 2x \Rightarrow dx = \frac{du}{2x}$$

$$\text{Rewrite } \int x e^u \frac{du}{2x} = \frac{1}{2} \int e^u du$$

$$= \frac{1}{2} e^u + c$$

$$\text{Back substitute : } \frac{1}{2} e^{x^2} + c ;$$

• Example

④ (i)

$$\int x (x^2 + 1)^{20} dx$$

Substitute $u = (x^2 + 1)$

$$\text{differentiate } \frac{du}{dx} = 2x \Rightarrow dx = \frac{du}{2x}$$

~~Rewrite $\int x e^x$~~

$$\text{Rewrite the problem } \int x \cdot u^{20} \cdot \frac{du}{2x}$$

$$= \int \frac{u^{20}}{2} du$$

$$= \frac{1}{2} x \frac{u^{20+1}}{20+1} + c$$

$$= \frac{1}{2} x \frac{u^{21}}{21} + c$$

$$\text{Back substitute : } \frac{u^{21}}{42} + c$$

$$= \frac{(x^2 + 1)^{21}}{42} + c ; c \text{ is an arb. Constant.}$$

$$(3) \int \frac{x}{\sqrt{x+1}} dx$$

Substitute $u = x+1$

differentiate $\frac{du}{dx} = 1 \Rightarrow dx = \frac{du}{1}$

Rewrite the problem $\int \frac{x}{\sqrt{u}} \cdot du$

$$= \int \frac{x}{u^{1/2}} \cdot du$$

$$= \int \frac{u-1}{\sqrt{u}} du$$

$$= \int \frac{u}{\sqrt{u}} du - \int \frac{1}{\sqrt{u}} du$$

$$= \int \frac{u}{u^{1/2}} du - \int \frac{1}{u^{1/2}} du$$

$$= \int \frac{u}{u^{1/2}} du - \int \frac{1}{u^{1/2}} du$$

$$= \int \frac{u^{1+1}}{u^{1/2+1}} du - \int \frac{1}{u^{1/2}} du \quad \leftarrow \text{exponents rules,}$$

$$= \int \frac{1}{u^{-1/2}} du - \int u^{-1/2} du$$

$$= \int u^{1/2} du - \int u^{-1/2} du$$

$$= \frac{u^{1/2+1}}{\frac{1}{2}+1} - \frac{u^{-1/2+1}}{-\frac{1}{2}+1} + C$$

$$= \frac{2}{3} u^{3/2} - 2 u^{1/2} + C$$

Back substitute : $\frac{2}{3} (x+1)^{3/2} - 2(x+1)^{1/2} + C$;

$$\begin{aligned}
 \text{(iii)} \quad \int e^{ax+b} dx &= \int e^u \cdot \frac{1}{a} du \\
 &= \frac{1}{a} \int e^u du \\
 &= \frac{1}{a} e^u + C \rightarrow \frac{1}{a} e^{ax+b} + C ;
 \end{aligned}$$

$$\begin{aligned}
 \text{(iv)} \quad \int (ax+b)^n dx &= \int u^n \cdot \frac{1}{a} du \\
 &= \frac{1}{a} \int u^n du \\
 &= \frac{1}{a} \times \frac{u^{n+1}}{n+1} + C \\
 &= \frac{1}{a} \frac{(ax+b)^{n+1}}{n+1} + C ;
 \end{aligned}$$

$$\begin{aligned}
 \text{(v)} \quad \int \frac{\ln x}{x} dx &= \int \frac{u}{x} \cdot x du \\
 &= \int u du \\
 &= \frac{u^{1+1}}{1+1} + C \\
 &= \frac{u^2}{2} + C ; \\
 &= \frac{(\ln x)^2}{2} + C ;
 \end{aligned}$$

* Integration by parts (IBP) : what & when

$$\boxed{\int u dv = uv - \int v du}$$

$$\text{e.g.} \rightarrow x e^x dx = x e^x - \int e^x dx$$

$$= x e^x - e^x + C ; C \text{ is an arb constant.}$$

✓ How to choose u and dv (LIATE rule)
Heuristic (LIATE) :

✓ Log > Inverse trig > Algebraic > Trig > Exponential

$$\text{e.g.} \rightarrow \int \underbrace{\ln x}_u \underbrace{dx}_{dv}$$

$$u = \ln x$$

$$\frac{du}{dx} = \frac{1}{x}$$

$$dv = dx$$

$$\int dv = \int dx$$

$$\bullet v = x$$

$$\bullet du = \frac{1}{x} dx$$

$$\int \ln x dx = \ln x \times x - \int x \cdot \frac{1}{x} dx$$

$$= x \ln x - \int dx$$

$$= \underline{x \ln x - x + C ; C \text{ is an arb constant.}}$$

$$\textcircled{1} \int \underbrace{x}_u \underbrace{(\sin x) dx}_{dv}$$

$$u = x$$

$$\frac{du}{dx} = 1$$

$$\int dv = \int \sin x dx$$

$$\bullet v = -\cos x$$

$$\bullet du = dx$$

$$\int x \sin x dx = x (-\cos x) - \int (-\cos x) dx$$

$$= -x \cos x + \int \cos x dx$$

$$= -$$

② ~~$\int x e^x dx$~~

$$\int x \underbrace{\ln x}_u dx$$

$$= \frac{x^2 \ln x}{2} - \frac{1}{2} \int x dx$$

$$= \frac{x^2 \ln x}{2} - \frac{1}{2} \cdot \frac{x^2}{2} + C$$

$$= \frac{x^2 \ln x}{2} - \frac{x^2}{4} + C$$

$$u = \ln x$$

$$\frac{du}{dx} = \frac{1}{x}$$

$$\frac{du}{1} = \frac{1}{2} dx$$

$$\bullet du = \frac{1}{2} dx$$

$$dv = x dx$$

$$\int dv = \int x dx$$

$$v = \frac{x^2}{2}$$

$$\bullet v = \frac{x^2}{2}$$

③ $\int x^2 e^x dx$

$$= x^2 \cdot e^x - \int e^x \cdot 2x dx$$

$$= x^2 e^x - 2 \int x e^x dx$$

$$u = x^2$$

$$\frac{du}{dx} = 2x$$

$$\bullet du = 2x dx$$

$$dv = e^x dx$$

$$\int dv = \int e^x dx$$

$$\bullet v = e^x$$

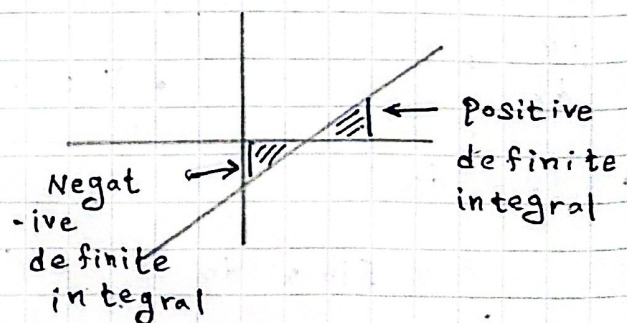
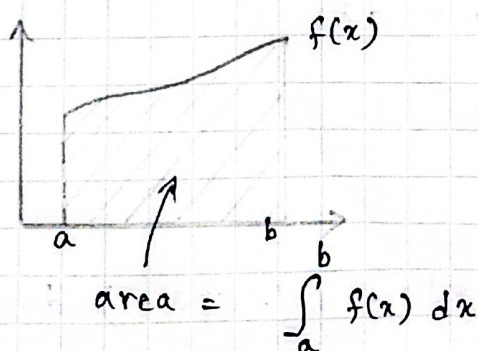
$$= x^2 e^x - 2 [x e^x - e^x] + C$$

$$= x^2 e^x - 2x e^x + 2e^x + C$$

* Definite Integral : Definition.

Fundamental Theorem of Calculus,

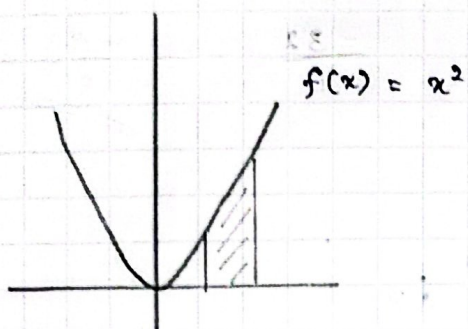
$$\int_a^b f(x) dx = F(b) - F(a)$$



Where, $F(x)$ is any antiderivative of $f(x)$

upper limit
 $\int_1^2 x^2 dx$
 lower limit

U.L
 $\int_1^2 x^2 dx$
 L.L



$$f(x) = x^2$$

$$\int f(x) dx = \int x^2 dx$$

$$f(x) = \frac{x^3}{3} + C$$

C is an
arb. constant

Example,

$$\textcircled{1} \int_1^2 x^2 dx = \left. \frac{x^3}{3} \right|_1^2$$

$$= \frac{2^3}{3} - \frac{1^3}{3} = \frac{8}{3} - \frac{1}{3} \rightarrow \frac{7}{3}$$

$$\textcircled{2} \int_0^{-1} x^6 dx = \left. \frac{x^7}{7} \right|_0^{-1}$$

$$= \frac{-1^7}{7} - \frac{0^7}{7} = \frac{-1}{7} - \frac{0}{7} = \frac{-1}{7}$$

$$\textcircled{3} \int_1^3 (x^2 - 3x + 1) dx = \left(\frac{x^3}{3} - \frac{3x^2}{2} + x \right) \Big|_1^3$$

$$= \left(\frac{3^3}{3} - \frac{3 \cdot 3^2}{2} + 3 \right) - \left(\frac{1^3}{3} - \frac{3 \cdot 1^2}{2} + 1 \right)$$

$$= \left(\frac{27}{3} - \frac{27}{2} + \frac{3}{1} \right) - \left(\frac{1}{3} - \frac{3}{2} + \frac{1}{1} \right)$$

$$= \left(\frac{18 - 27 + 6}{2} \right) - \left(\frac{2 - 9 + 6}{6} \right)$$

$$= \frac{-3}{2} - \left(\frac{-1}{6} \right)$$

$$= \frac{-3}{2} + \frac{1}{6} = \frac{-9 + 1}{6}$$

$$= \frac{-8}{6} \rightarrow \underline{\underline{\frac{-4}{3}}}$$

$$\textcircled{4} \int_1^9 (\sqrt{x+1}) dx = \int_1^9 (x^{\frac{1}{2}} + 1) dx$$

$$= \frac{x^{\frac{3}{2}}}{\frac{3}{2}} + x \Big|_1^9$$

$$= \left(\frac{9^{\frac{3}{2}}}{\frac{3}{2}} + 9 \right) - \left(\frac{1^{\frac{3}{2}}}{\frac{3}{2}} + 1 \right)$$

$$= \left(\frac{3 \cdot 2 \cdot 3}{\frac{3}{2}} + 9 \right) - \left(\frac{1^{\frac{3}{2}}}{\frac{3}{2}} + 1 \right)$$

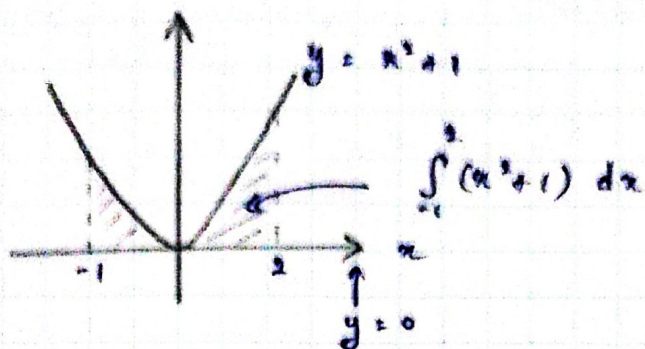
$$= \left(\frac{27}{\frac{3}{2}} + 9 \right) - \left(\frac{1}{\frac{3}{2}} + 1 \right)$$

$$= \left(\frac{27 \times 2}{3} + 9 \right) - \left(\frac{1 \times 2}{3} + 1 \right)$$

$$= \left(\frac{54}{3} + \frac{9}{1} \right) - \left(\frac{2}{3} + 1 \right)$$

$$= \left(\frac{54 + 27}{3} \right) - \left(\frac{2 + 3}{3} \right)$$

$$= \frac{81}{3} - \frac{5}{3} \rightarrow \frac{76}{3} \rightarrow \underline{\underline{\frac{76}{3}}}$$



$$\begin{aligned}
 \textcircled{1} \quad \int_{-1}^2 x^2 + 1 \, dx &= \int_{-1}^2 x^2 \, dx + \int_{-1}^2 1 \, dx \\
 &= \left. \frac{x^3}{3} \right|_{-1}^2 + \left. x \right|_{-1}^2 \\
 &= \left(\frac{2^3}{3} \right) - \left(\frac{(-1)^3}{3} \right) + (2 - (-1)) \\
 &= \frac{8 + 1}{3} + 3 \\
 &= 3 + 3 \rightarrow \underline{\underline{6}}
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{2} \quad \int_0^4 \sqrt{x} \, dx &= \left. \frac{x^{3/2}}{3/2} \right|_0^4 \\
 &= \frac{2}{3} \left[4^{3/2} - 0^{3/2} \right] \\
 &= \frac{2}{3} \left(2^{2 \times \frac{3}{2}} - 0 \right) \rightarrow \frac{2}{3} \times 8 \\
 &= \underline{\underline{\frac{16}{3}}}
 \end{aligned}$$